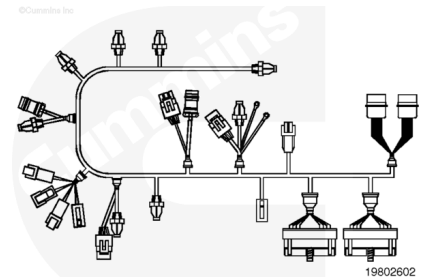


Trainings ▾

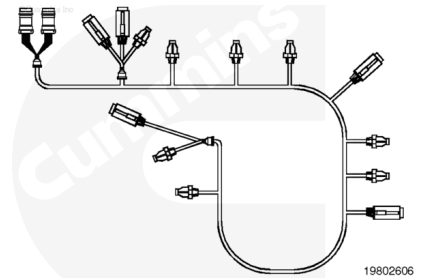
General Information

CENSE™-equipped engines use three separate wiring harnesses:

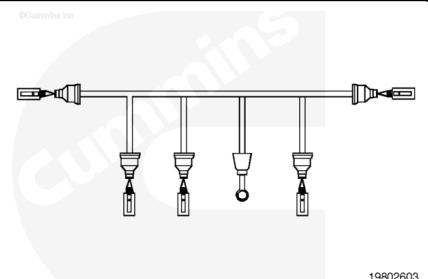
- CENSE™ harness, main/left bank sensors



- CENSE™ harness, right bank sensors



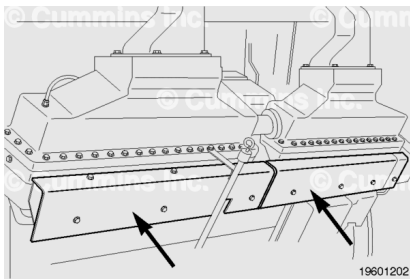
- J1939 communication harness (backbone harness).



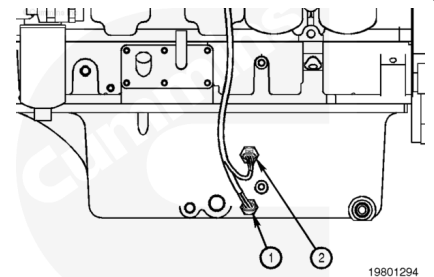
Remove

Left Bank

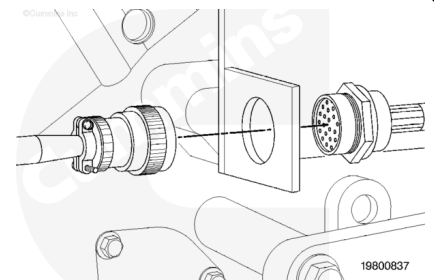
Remove the left bank harness covers.



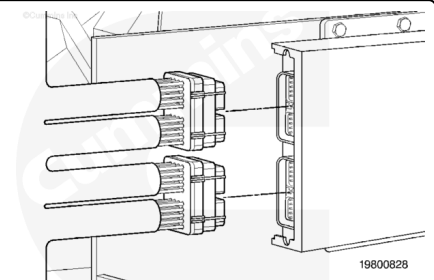
Disconnect the main/left bank CENSE™ harness from the oil temperature sensor (1).



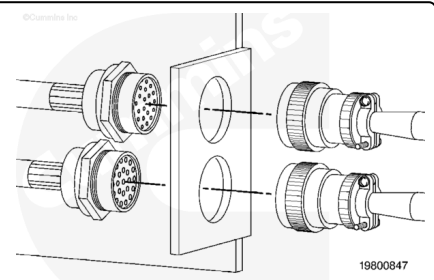
Disconnect the 23-pin OEM harness connector.



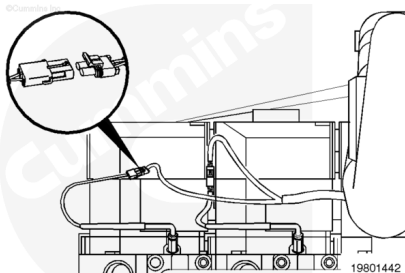
Disconnect the 40-pin A and B ECM connectors.



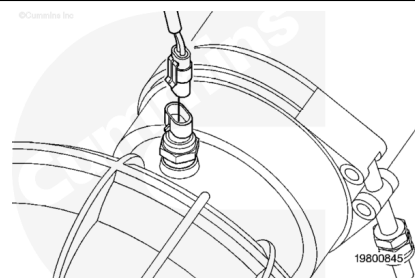
Disconnect the 23-pin and the 33-pin right bank sensor harness connectors.



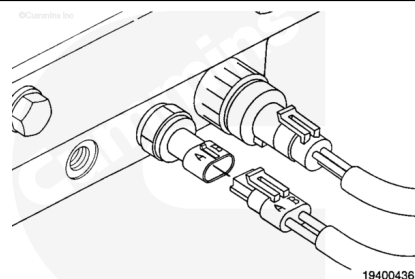
Disconnect all of the left bank exhaust temperature sensors.



Disconnect the left bank rear turbocharger compressor inlet temperature sensor.

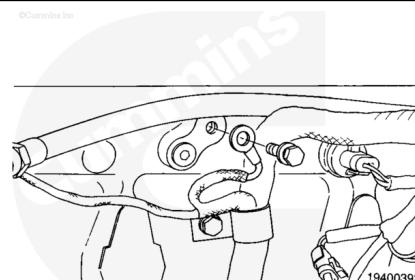


Disconnect the 2-pin left bank rear intake manifold temperature sensor.



Disconnect the engine block ground from the engine block. There may be one or more ring terminals to disconnect.

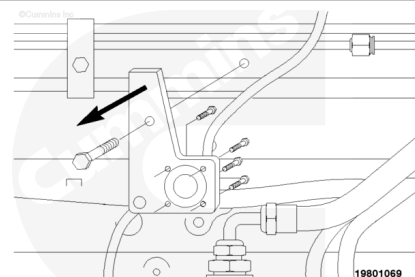
Note : Some harnesses have one common ground.



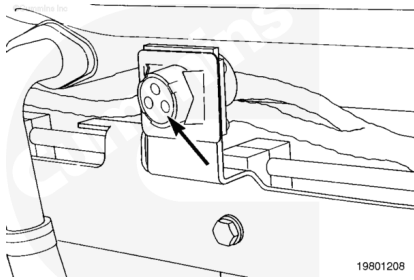
Disconnect the QSK45/60 2-way interharness connector (data link) from the engine control system harness.

- Remove the capscrews.
- Slide the data link connector out of the mounting slot in the bracket.

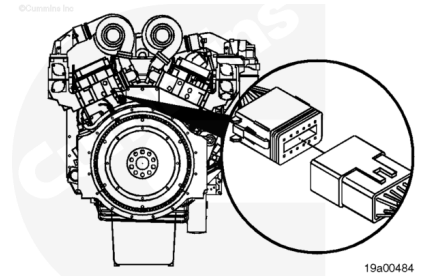
Note : The bracket may differ from the illustration.



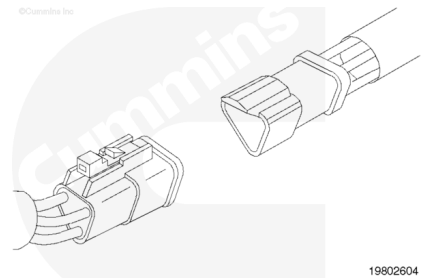
Disconnect the RS232 3-pin data link connector from the support bracket.



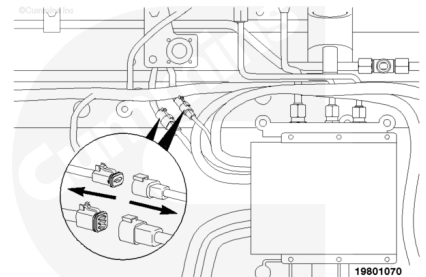
Disconnect the warning lamps connector from the QSK45/60 lamp connector.



Disconnect the 3-pin Deutsch™ J1939 data link from the communication harness (backbone).



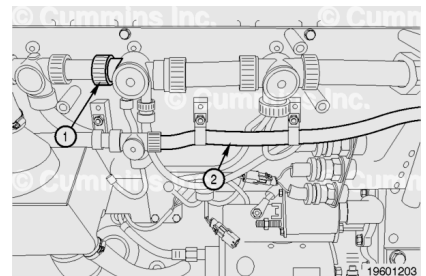
Disconnect the 6-pin Deutsch™ connectors from the QSK45/60 6-way interharness connectors.



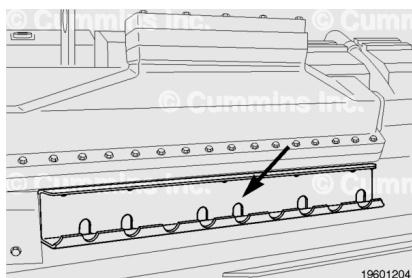
Remove the main/left bank harness (1) and backbone harness (2) from the support bracket.

Cut all of the nylon wire ties from the CENSE™ main/left bank harness.

Remove the harnesses from the support brackets.

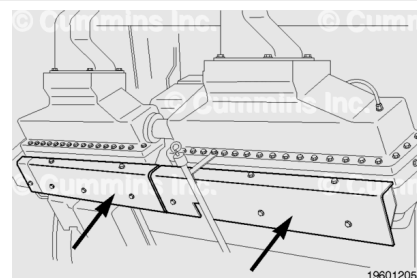


Remove the harness support brackets.

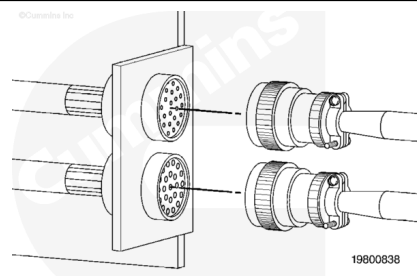


Right Bank

Remove the right bank harness covers.

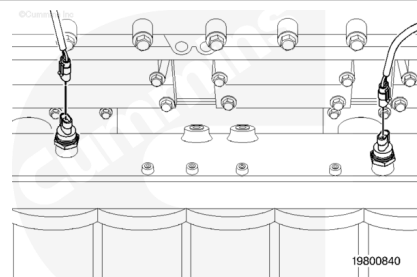


Disconnect the 23-pin and the 33-pin Deutsch™ connectors on the right bank sensor harness.

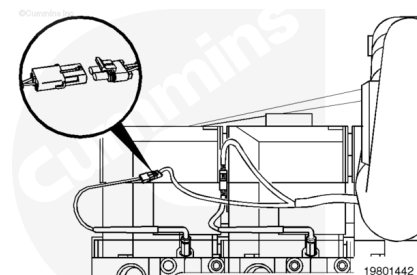


Disconnect the pre-filter lubricating oil pressure sensor and the post-filter lubricating oil pressure sensor.

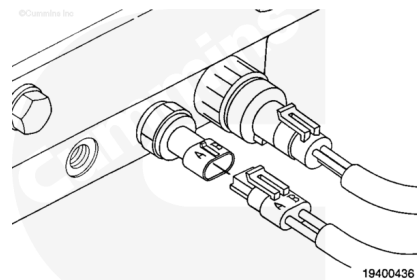
Note : This step only applies for engines with pre-filter and post-filter lubricating oil pressure sensors on the right bank.



Disconnect all of the right bank exhaust temperature sensors.

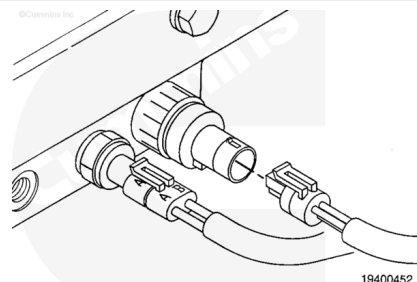


Disconnect the right bank front and right bank rear intake manifold temperature sensor 2-pin connectors.



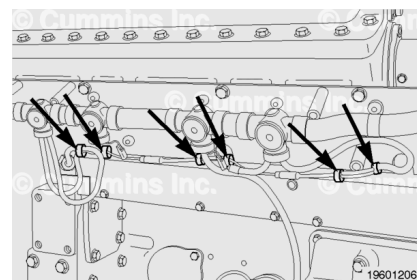
Disconnect the right bank boost pressure sensor 3-pin connector.

The right bank boost pressure sensor is located in the right bank rear intake manifold.

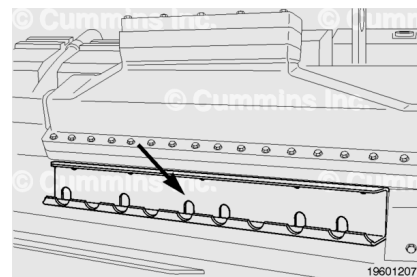


Remove the harness t-pieces and p-clips from the support brackets.

Cut all of the nylon wire ties from the CENSE™ right bank harness.



Remove the right bank harness support brackets.



Install

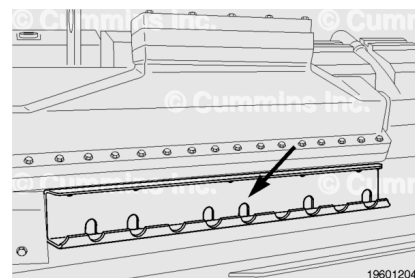
Left Bank

Attach the harness support brackets to the intake manifolds in the locations shown.

Apply Loctite® 243, or equivalent, to the capscrews.

Insert and tighten the capscrews.

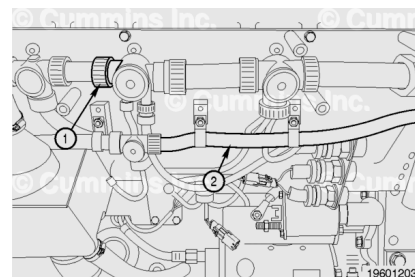
Torque Value: 23 n•m [204 in-lb]



Attach the main/left bank harness (1) and backbone harness (2) to the support bracket. Insert the mounting nuts and tighten.

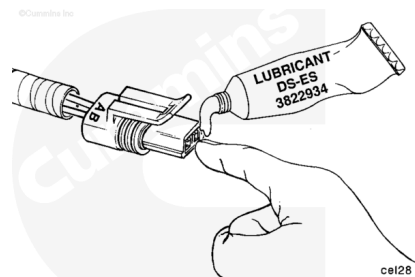
Torque Value: 10 n•m [89 in-lb]

Use nylon wire ties, where required, to secure the harness to the engine.



⚠ CAUTION ⚠

Use only Cummins®-recommended lubricant DS-ES, Part Number 3822934. Other lubricants, such as lubricating oil or grease, in the connectors can cause ECM damage, poor engine performance, or premature connector wear.



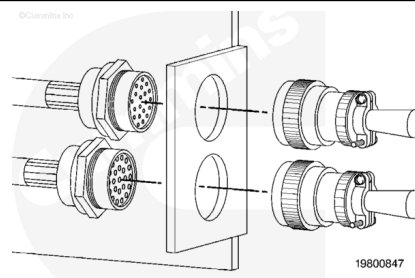
Apply a small amount of lubricant to the connector terminals. Before installing, fill the entire connector cavity with lubricant.

Lubricate all harness connectors.

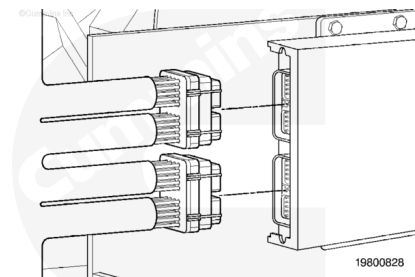
Install the 23-pin and the 33-pin right bank sensor harness connectors on the mounting plate.

Torque Value: 20 n•m [177 in-lb]

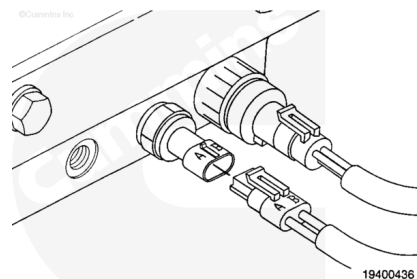
Connect the 23-pin and the 33-pin connectors.



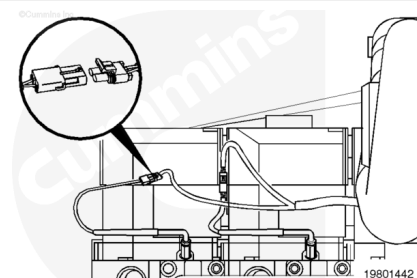
Connect the 40-pin A and B ECM connectors.



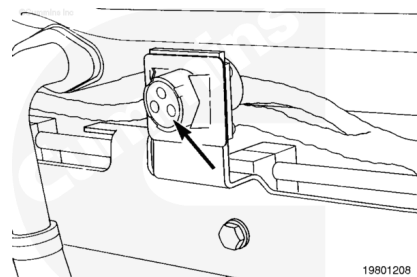
Connect the 2-pin left bank rear intake manifold temperature sensor.



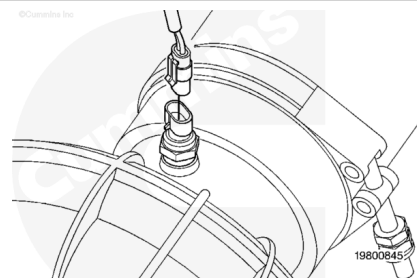
Connect all of the left bank exhaust temperature sensors.
Refer to Procedure 019-013 (Exhaust Temperature Sensor) in Section 19. (</qs3/pubsys2/xml/en/procedures/81/81-019-013.html>)



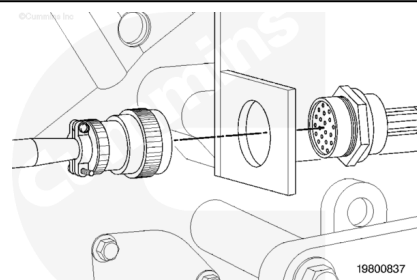
Install the RS232 3-pin data link connector into the support bracket.



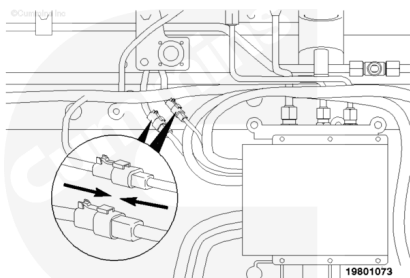
Connect the left bank rear turbocharger compression inlet temperature sensor.



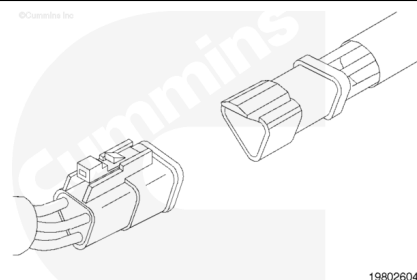
Connect the 23-pin OEM harness connector.



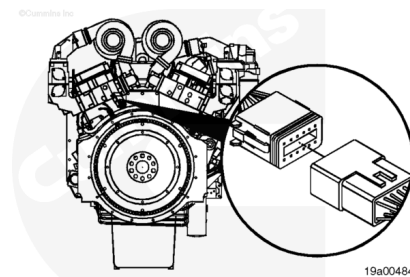
Connect the 6-pin Deutsch™ connectors to the QSK45/60 6-way interharness connectors.



Connect the 3-pin Deutsch™ J1939 data link to the communication harness (backbone).



Connect the warning lamp connector to the QSK45/60 lamp connector.

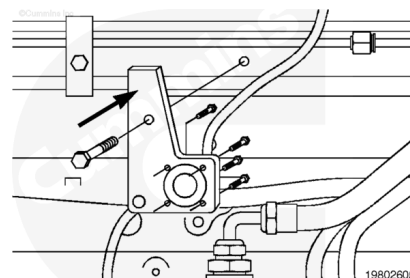


Connect the QSK45/60 2-way interharness connector (data link).

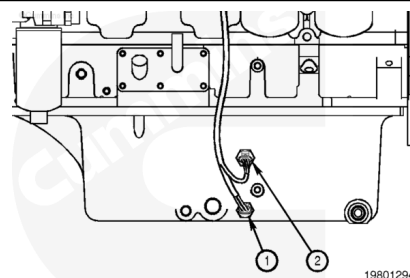
- Slide the connector into the support bracket.
- Secure the connector with screws.

Torque Value: 1.2 n·m [11 in-lb]

Note : The bracket may differ from the illustration.

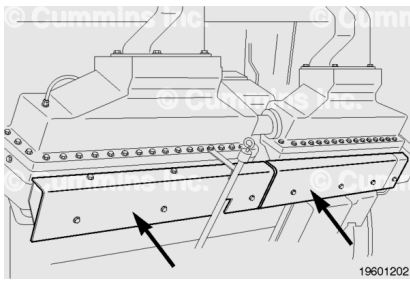


Connect the oil temperature sensor (1).



Attach the left bank harness covers.

Torque Value: 45 n·m [33 ft-lb]



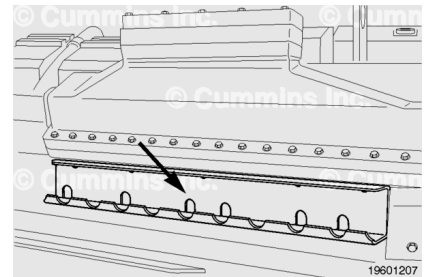
Right Bank

Attach the harness support brackets to the intake manifolds in the locations shown.

Apply Loctite® 243, or equivalent, to the capscrews.

Insert and tighten the capscrews.

Torque Value: 23 n•m [204 in-lb]

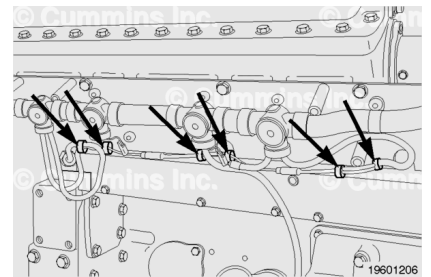


Attach the harness t-pieces and p-clips to the support brackets.

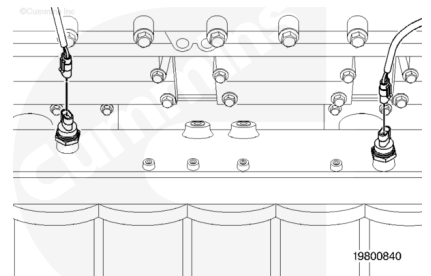
Torque Value: 10 n•m [89 in-lb]

Insert the mounting nuts and tighten.

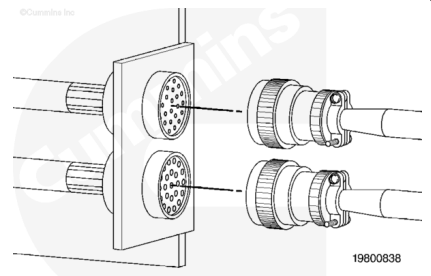
Use nylon wire ties, where required, to secure the harness to the engine.



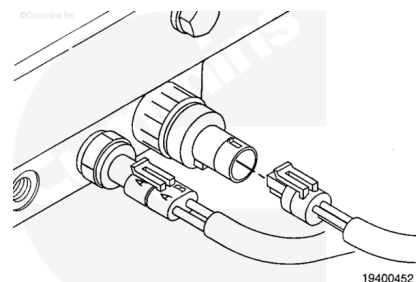
Connect the pre-filter lubricating oil pressure sensor and the post-filter lubricating oil pressure sensor.



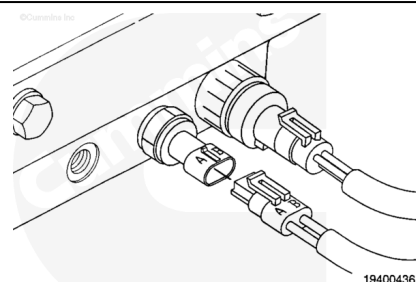
Connect the 23-pin and the 33-pin Deutsch™ connectors on the right bank sensor harness.



Connect the right bank boost pressure sensor 3-pin connector, located in the right bank rear intake manifold.

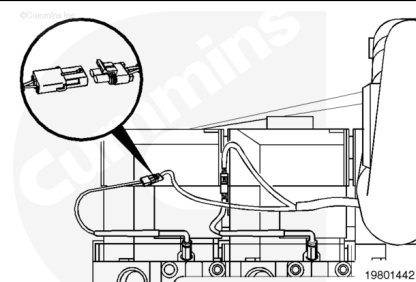


Connect the right bank front and the right bank rear intake manifold temperature sensor 2-pin connectors.



Connect the right bank exhaust temperature sensors.

Refer to Procedure 019-013 (Exhaust Temperature Sensor) in Section 19. (</qs3/pubsys2/xml/en/procedures/81/81-019-013.html>)



Attach the right bank harness covers.

Torque Value: 45 n•m [33 ft•lb]

